Aditi Roy Adri

aditiroyadri121@gmail.com | +880 1315841499 | Dhaka, Bangladesh
github.com/Aditi-Adri | linkedin.com/in/aditi-roy-adri
aditi-adri.github.io/AditiRoyAdri

Summary

Computer Science and Engineering graduate from BRAC University with a strong foundation in software development, machine learning, and data analysis. Experienced in developing web applications, implementing deep learning models, and conducting research. Interested in applying technical skills to solve real-world problems while continuing to learn and grow as a technology professional.

Education

- **BRAC University, Dhaka** *2023 – 2026*
  Bachelor of Science in Computer Science and Engineering **CGPA:** 3.83/4.00
- **Holy Cross College, Dhaka** *2021*
  Higher Secondary Certificate (HSC) **GPA:** 5.00/5.00
- **Narayanganj High School and College, Narayanganj** *2019*
  Secondary School Certificate (SSC) **GPA:** 5.00/5.00

Technical Skills

- **Programming Languages:** Python, C++, C, Java, PHP, Assembly (Emu8086), Bash
- **Machine Learning & Computer Vision:** Deep Learning, CNNs, Image Segmentation, Object Detection, Transfer Learning, Model Evaluation, YOLO, U-Net
- **Web Technologies:** HTML5, CSS3, JavaScript, React.js
- **Databases:** MySQL, PostgreSQL
- **Libraries & Frameworks:** PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, Jupyter Notebook
- **Graphics & Simulation:** OpenGL (PyOpenGL), Cisco Packet Tracer
- **Tools & Platforms:** Git, GitHub, LaTeX, Linux, VS Code

Projects

- **CARDD – Car Damage Detection with YOLO** *July 2026*
  Trained and benchmarked YOLO11-L against YOLOv8-L on the CarDD dataset (4,000 images, 8,740 annotated boxes across 6 damage classes: scratches, dents, cracks, broken lamps, shattered glass, flat tires). YOLO11-L achieved **mAP@50–95 of 0.557**.
  Tech: Python, YOLO11, YOLOv8, PyTorch, Object Detection GitHub
- **Pet Image Segmentation & Breed Classification (U-Net)** *May 2026*
  Deep learning project using U-Net and Attention U-Net architectures for binary pet image segmentation and 37-breed classification on the Oxford-IIIT Pet Dataset. Achieved **∼88% IoU** with a pretrained ResNet-34 encoder. Implemented data augmentation, combined loss functions, and ablation studies.
  Tech: Python, PyTorch, Jupyter Notebook GitHub
- **Humanitarian Classification of Disaster Tweets** *April 2026*
  Benchmarked ten models on the HumAID dataset (76,484 tweets, 19 disasters, 10 humanitarian categories with a 59.6:1 class imbalance). **BERT reached 0.768 macro F1**; TF-IDF logistic regression followed at 0.715, outperforming every recurrent model tested.
  Tech: Python, BERT, TF-IDF, Scikit-Learn, NLP GitHub
- **SaveLife – SOS Emergency Response App** *March 2026*
  Emergency response platform with one-tap SOS alerts and GPS geolocation, a live responder dashboard, and role-based JWT authentication. React/Vite frontend, Node.js/Express API, PostgreSQL,

deployed on Render.
Tech: React, Vite, Node.js, Express, PostgreSQL, JWT GitHub | Live Demo

- **Roaming Sonic – Travel & Tour Management Platform** *Jan 2026*
  Full-stack MERN web application for Bangladesh tourism featuring tour packages, hotel bookings, bus tickets, guide services, AI chatbot, and referral rewards system. Live-deployed on Vercel.
  Tech: MongoDB, Express.js, React.js, Node.js GitHub | Live Demo

- **Metadata Journal System** *Oct 2025*
  OS-level project implementing file metadata management and journaling mechanisms for maintaining filesystem consistency.
  Tech: C GitHub

Certifications

- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI, Coursera
- **IBM Data Science** (6 courses) — IBM, Coursera: What is Data Science?; Tools for Data Science; Data Science Methodology; Python for Data Science, AI & Development; Python Project for Data Science; Databases and SQL for Data Science with Python
- **Foundations of Cybersecurity** — Google, Coursera
- **BRAC University Programming Contest** — BRAC University Computer Club

Research Experience

- **Undergraduate Thesis: TAM-Net for Underwater Segmentation** *2025 – 2026*
  BRAC University, Dhaka
  Developed a four-class underwater semantic segmentation dataset and proposed TAM-Net, a teacher-assisted masking framework with a small-object detail pathway, achieving 74.7% mean Intersection over Union (mIoU).
  Supervised by Md. Tanzim Reza (Senior Lecturer) and Dr. Md. Golam Rabiul Alam (Professor), Department of CSE, BRAC University.
- **Research Collaborator** *2026 – Present*
  BRAC University, Dhaka
  Collaborating with faculty from the Departments of Computer Science and Engineering and Law on interdisciplinary research. Contributions include software development, data curation, formal analysis, visualization, and manuscript writing.

Leadership & Activities

- **Women & Child Affairs Secretary**, Executive Committee — Bangladesh Youngster Social Organization (BYSO), Dhaka District Unit, for the 2026–27 session. Leadership role supporting women's and children's welfare initiatives.
- Competitive programming on Codeforces (500+ problems solved in Python).
- Participation in university-level and inter-university programming contests at BRAC University.
- Active volunteer in community and social-welfare activities.
- Vocalist for cultural clubs and events at college and university.

References

Upal Aditya Oikya, Ph.D.
Assistant Professor
School of Law
BRAC University, Dhaka, Bangladesh
+880 9638-464646, Ext. 2309
aditya.oikya@bracu.ac.bd

Ipshita Bonhi Upoma
Lecturer
Dept. of Computer Science & Engineering
School of Data and Sciences
BRAC University, Dhaka, Bangladesh
+880 1673-584606
ipshita.upoma@bracu.ac.bd